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17 November 1965
Superseding
MIL-I-27552A(USAF)
27 October 1961

MILITARY SPECIFICATION

INDICATOR, TEMPERATURE, THERMOCOUPLE, POTENTIOMETER TYPE, WHITE LIGHTED

1. SCOPE

1.1 Scope. This specification covers the requirements for white-lighted, potentiometer-type, self-balancing temperature indicators.

1.2 Classification. Indicators shall be of the following types, as specified (see 6.2):

EHU-15/A	Antireflective coating on cover glass per MIL-C-675
EHU-15A/A	High-efficiency antireflective coating on cover glass

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Method of
MIL-C-675	Coating of Glass Optical Elements (Anti-Reflection)
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5400	Electonic Equipment Aircraft, General Specification for
MIL-W-5846	Wire, Electrical, Chromel and/or Alumel, Thermocouple
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-M-26512	Maintainability Program Requirements for Aerospace Systems and Equipment
MIL-R-26667	Reliability and Longevity Requirements, Electronic Equipment, General Specification for
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking for Shipment and Storage

FSC 6685

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MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards Order of Precedence for the Selection of
MIL-STD-704	Electric Power, Aircraft Characteristics and Utilization of
MIL-STD-785	Requirements for Reliability Program (for Systems and Equipments)
MS3106	Connector, Plug, Electric, Straight
MS33558	Numerals and Letters, Aircraft Instrument
MS33585	Dial, Standard Form of Pointers, Dial, Standard Design of Aircraft Instrument
MS33586	Metals, Definition of Dissimilar
MS33639	Cases, Instrument, Clamp-Mounted, Aircraft
MS33678	Connector, Receptacle, Electric, Integral Mounting

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards

Circular 561	Reference Tables for Thermocouples

(Applications for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

Federal Aviation Agency

Technical Standard Order

TSO-C43	Temperature Indicators

(Application for copies should be addressed to the Federal Aviation Agency, Washington, D. C. 20553.)

3. REQUIREMENTS

3.1 Qualification. The indicator furnished under this specification shall be a product that has been tested, and has passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Preliminary list of electronic parts. A preliminary list of electronic parts shall be furnished directly to the procuring activity in accordance with MIL-E-5400 prior to ordering or fabricating parts for the production quantity. The parts shall also conform to MIL-E-5400. Nonstandard parts shall be approved by the procuring activity prior to fabrication of the completed item. Standard electronic parts shall be used whenever practicable. Nonstandard electronic parts shall not constitute more than

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50 percent of the total quantity of electronic parts used in the construction of the indicator.

3.4 Materials

3.4.1 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the indicator, except where magnetic materials are essential.

3.4.2 Nonferrous materials. Nonferrous materials shall be used for all parts of the indicator except where ferrous materials are essential.

3.4.3 Metals. Metals shall be of the corrosion-resistant type or shall be suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

3.4.4 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissmililar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.4.5 Toxic and corrosive fumes. The indicator shall be of such material and construction that fumes, when burned, or gases that combine with the atmosphere to form acid or corrosive alkali will not be liberated.

3.4.6 Protective treatment. When materials are used in the construction of the indicator that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.5 Design and construction. The design and construction of the indicator shall include all of the circuitry necessary for converting chromel-alumel thermocouple signals to temperature indication on the aircraft instrument panel, a dial conforming to figure 1, and a self-balancing potentiometer-type circuit for temperature measurements.

3.5.1 Power consumption. The indicator power consumption shall not exceed 10 va, exclusive of the lighting circuit.

3.5.2 Power supply. The indicator shall be designed to operate from a 115V 400-cycle, single-phase power supply conforming to MIL-STD-704 except that the voltage and frequency each may vary ±10 percent. Operation with surge voltages shall be as specified in 4.6.12.

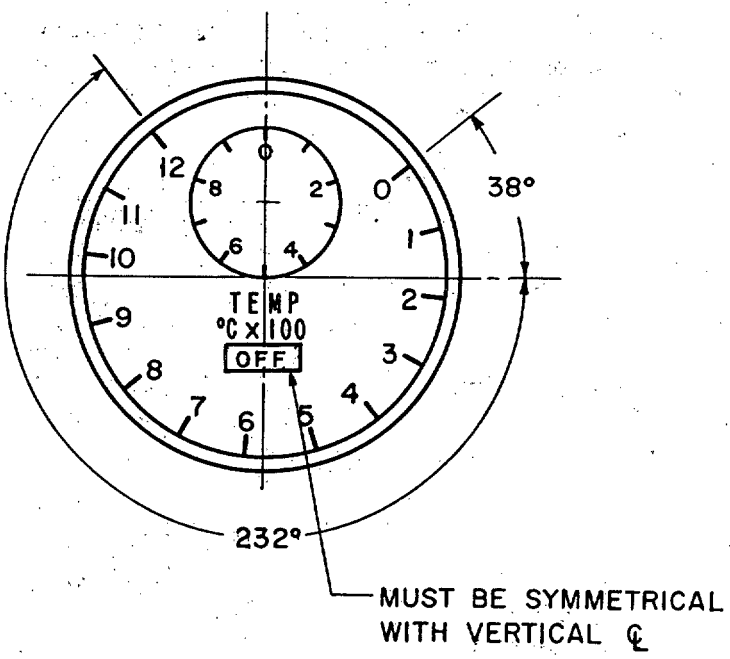
3.5.3 Temperature signals. The indicator shall be designed to operate on signals generated by chromel-alumel thermocouples. The values of the signals shall be as specified in the National Bureau of Standards Circular 561.

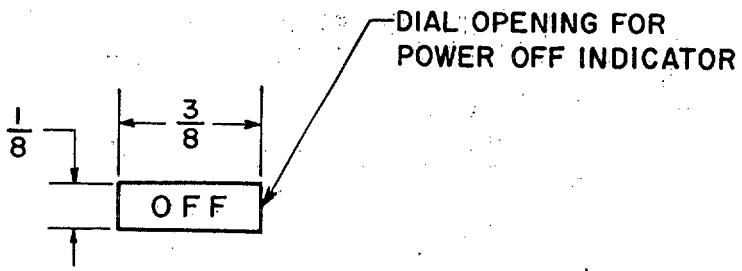
3.5.4 Reliability program. The contractor shall establish a reliability assurance program in accordance with MIL-STD-785.

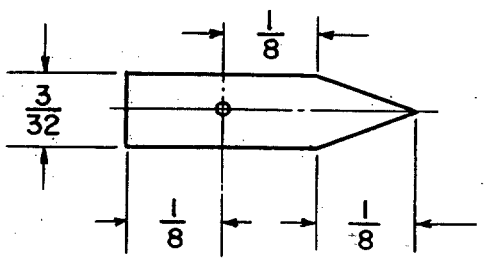
3.5.4.1 The indicator shall have a minimum mean-time-between-failures of 500 hours at a 90 percent confidence level during its service life.

3.5.5 Maintainability. Maintainability shall be in accordance with MIL-M-26512.

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SUBDIAL POINTER
SEE SECTION 3

DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED,
TOLERANCE: FRACTIONS ± 1/64
TOTAL SCALE ANGLE ± 1°

FIGURE 1. Dial

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3.5.6 TSO conformance. The indicator shall conform to the requirements of Federal Aviation Agency TSO-C43 for type II indicators.

3.5.7 Cover glass coating. The reflecting glass surface of the EHU-15/A indicator shall be finished with a reflection-reducing coating conforming to MIL-C-675. The EHU-15A/A indicator cover glass shall be finished with a high-efficiency coating which meets the following. The coatings shall withstand the environmental conditions specified herein.

Angle of Incidence	Wave Length Millimicrons	Percent Reflectance
0°	450 - 675	0.6 maximum
0°	425 - 700	0.5 average
30°	450 - 625	1.0 maximum
30°	425 - 700	0.5 average

3.6 Performance. The indicator shall be capable of meeting the following requirements:

a. Scale error not exceeding the tolerances specified in table I

b. Friction error not exceeding 2°C as viewed on the subdial

c. Position error not exceeding 1°C

d. Case leakage not exceeding 0.1 micron cubic foot per hour with a nitrogen-helium mixture and 1.0 micron cubic foot per hour with a 100 percent helium mixture

e. Warmup and pointer travel from 100° to 1,000°C within 6 seconds at room temperature and 10 seconds -55° and +100°C

f. Instantaneous indication change from 700° to 1,050°C in not more than 2 seconds at room temperature and 100°C and 3 seconds at -55°C

g. A maximum of 2° ±1°C indication change at points between 800° and 1,000°C when the power is increased or decreased 0.08 mv

h. Temperatures ranging from -65° to +100°C

i. Withstand applied vibration of 0.009 to 0.018 amplitude at frequencies ranging from 5 to 50 cps

j. Withstand an alternate input of -3 and +52 mv for a period of 30 minutes at each condition for 8 cycles with the ambient temperature at 50°C

k. Indication change not exceeding 2°C when a lead resistance change from 5 to 200 ohms is made at specified test points between 500° and 1,000°C

l. Indicator steady state change not exceeding 2°C when either the chromel or alumel conductor is grounded separately at any point to the ground side (low potential side) of the 115V 400-cycle power supply

m. Withstand the effects of beat frequencies and stray a-c signals as specified in 4.6.17

n. Withstand pressures ranging from 30 inches Hg down to 0.309 inches Hg, approximately sea level to 100,000 feet

o. Withstand magnet effect when positioned 5 ±1/2 inch from a magnet which produces a field intensity of 0.17 to 0.19 oersted at the indicator

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TABLE I

Indicator Tolerances

Test Points °C	Room Temp Scale Error ±°C	Low Temp Scale Error ±°C	High Temp Scale Error ±°C	Scale Error After Reliability ±°C	Position Error ±°C	Friction Error ±°C
100	10.0	15.0	15.0	15.0	---	2.0
200	5.0	10.0	10.0	10.0	---	2.0
400	5.0	10.0	10.0	10.0	---	2.0
500	5.0	10.0	10.0	10.0	---	2.0
600	5.0	10.0	10.0	10.0	---	2.0
700	5.0	10.0	10.0	10.0	---	2.0
800	5.0	10.0	10.0	10.0	1.0	2.0
900	5.0	10.0	10.0	10.0	1.0	2.0
1,000	5.0	10.0	10.0	10.0	1.0	2.0
1,100	5.0	10.0	10.0	10.0	---	2.0
1,200	10.0	15.0	15.0	20.0	---	2.0

p. Withstand frequency variations ranging from 360 to 440 cps and voltage variations from 104 to 126V

q. Withstand 8 cycles of immersion in tap water maintained at 85° ±2°C and 5° ±2°C

r. Power OFF warning shall withstand 20,000 cycles of operation at 15 ±2 cpm

s. Pointer cycling from 100° to 1,000°C and return to 100°C for 100,000 cycles at a rate of 2.5 ±0.5 cpm

t. Exposure to salt sea atmosphere for a period of 50 hours

u. Withstand voltage surges up to 195 ±5V.

3.6.1 Radio noise interference. The indicator shall suppress internally radiated and conducted radio noise within the limits specified in MIL-I-6181.

3.7 Case. The case shall conform to MS33639 for the 2-inch size, shall be finished in durable dull black, and shall be hermetically sealed. The seal shall be accomplished in a manner that will not be dependent upon materials that might be affected by the action of any atmosphere to which the case will be subjected. The front edge of the case shall not extend beyond the outer surface of the cover glass by more than 1/32 inch. The case length shall not exceed 6.0 inches exclusive of the electrical receptacle. The design of the case shall be such that the internal mechanism may be removed from the case, replaced, and the case resealed without the use of special tools, unless approved by the procuring activity.

3.7.1 Case grounding. The case backplate shall be connected to the external power ground by means of a ground lug permanently attached to the inner surface

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of the backplate. A ground wire shall be used to connect the ground lug to pin F of the electrical receptacle.

3.7.2 Filling medium. The filling medium shall be of at least 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The filling medium shall be either 100 percent helium or a mixture of 88 percent to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium in the case shall be approximately 1 atmosphere or 1/2 atmosphere, as required. Where practicable, the filling medium shall be 100 percent helium.

3.7.3 Filler tube. A tube or other facilities for evacuating and filling the case shall be provided at the rear of the case and shall be recessed beneath the level of the backplate surface.

3.7.4 Nameplate. A nameplate shall be provided on the rear of the case. The information shall conform to MIL-STD-130, where applicable.

3.8 Dials. The dials shall be linear and shall consist of a main dial and a subdial in accordance with figure 1. The subdial shall be 5/8 ±1/64 inch in diameter and shall have scale graduations on the main dial plate linearly spaced within 0.5 of an angular degree. The subdial may be fabricated on a counter-bored surface in the main dial plate, or on a recessed disc beneath the level of the main dial plate provided the reading angle for the subdial is, at minimum, equal to that of the main dial. The main dial scale shall be not less than 1.75 inches in diameter, as measured across the outer ends of the graduations, and shall have an arc of 270° ±1°.

3.8.1 Lighting. The dials shall be integrally lighted in accordance with MIL-L-27160.

3.8.2 Dial pointers. The main dial pointer shall conform substantially to MS33585-8 except for deviations required for internal lighting. The main dial pointer shall be lighted a minimum length of 0.60 inch, as measured from the tip of the pointer. The subdial pointer shall be lighted at least from tip to hub. Both pointers shall be integrally lighted in accordance with MIL-L-27160. An illuminated subdial pointer configuration on a black rotating disc may be used in lieu of a subdial pointer provided the lighting meets the requirements of MIL-L-27160.

3.8.2.1 Pointer overswing. Pointer overswing shall be 3° ±1° beyond each end of the dial scale. Pointer stops shall be used to limit pointer travel as specified herein.

3.8.3 Dial to cover glass distance. The maximum distance between the inner surface of the cover glass and the indicator dial shall be 0.188 inch.

3.8.4 Numerals and letters. Dial numerals and letters shall conform to MS33558.

3.9 Power-off warning. The indicator shall incorporate a POWER-OFF warning as shown on figure 1. The warning shall display the word OFF in black letters, color No. 17038 of FED-STD-595, on a conspicuous red background, Lawter Chemical, Incorporated daylight fluorescent red, color No. 34, or equal. The warning shall be lighted to a brightness level of 2 ±1 foot lambert and shall remain in view when power is not applied. When power is applied, the warning shall disappear from view within 1 second and

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shall be replaced by a dull black surface. The warning shall operate from the indicator power supply.

3.10 Thermocouple cold junction compensation. The indicator shall incorporate temperature compensation for the thermocouple cold junction within the indicator. Temperature control at the cold junction shall not be used. The compensation shall be such that the input temperature signal to the indicator circuit will be independent of the temperature changes at the cold junction.

3.11 Reference voltage. If reference voltage is required for the self-balancing potentiometer, it shall be derived from the indicator power supply. Standard cells shall not be used.

3.12 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-D-70327 shall govern the manufacturer's part numbers and changes thereto.

3.13 Electrical receptacle. The indicator shall be provided with an MS33678C-18-9P electrical receptacle, except that the receptacle shall be hermetically sealed. Pins A and D of the receptacle shall be made of alumel and chromel materials, respectively.

3.13.1 Wiring. The receptacle shall be wired as follows:

PIN	EXTERNAL CONNECTION
A	Alumel conductor
B	Open
C	115V 400 cps input
D	Chromel conductor
E	5V ac or dc input for lighting
F	115V 400 cps and 5V ground
G	Open

3.14 Weight. The weight of the indicator shall not exceed 1.5 pounds.

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the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25° C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Attitude. Unless otherwise specified, the indicator shall be tested in the normal operating position with the plane of the dial vertical.

4.3.3 Power supply. Unless otherwise specified, the indicator shall be tested with 115 ±2V, single-phase and 400 ±5 cps ac, with 4.5 to ±0.5V ac or dc applied to the lighting circuit.

4.3.4 During testing of the indicator, the power OFF warning shall be observed for proper operation.

4.3.5 Integral lighting. The integral lighting tests shall be conducted in the sequence specified and in accordance with the methods specified in MIL-L-27160. The background of the power OFF warning shall be lighted to a brightness level of 2 ±1 foot lambert.

4.4 Qualification testing

4.4.1 Qualification test samples. The qualification test samples shall consist of eight indicators representative of the production equipment. The samples shall be identified with the manufacturer's own part number and such other information as required by the procuring activity. Five of the indicators shall be used for reliability testing (see 4.6.23) and the other three indicators shall be subjected to all tests specified under 4.6 except reliability. In addition, three POWER-OFF warnings not previously endurance cycled or otherwise damaged shall be submitted complete and suitable for cycling tests for qualification approval.

4.4.1.1 When the tests are conducted at a location other than the laboratory of the qualifying activity, the test samples that were tested shall be furnished to that activity in the "as is" condition.

4.4.2 Qualification tests. The qualification tests shall consist of all the tests specified under 4.6.

4.5 Acceptance tests. Acceptance tests shall consist of:

a. Individual tests

b. Sampling tests.

4.5.1 Individual tests. Each indicator shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Individual tests of MIL-L-27160

c. Scale error and friction

d. Position error

e. Sealing.

4.5.2 Sampling Tests

4.5.2.1 Sampling plan A. One indicator shall be selected at random from each 100 or less produced on the contract or order and subjected to the following tests as described under 4.6:

a. Individual tests

b. Sampling plan A tests of MIL-L-27160

c. Warmup time

d. Response time

e. Sensitivity

f. Low temperature and compensator lag

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3.15 Markings

3.15.1 Dimensions and colors. The dimensions and colors of the dial and pointer markings shall be in accordance with table II. The colors shall conform to FED-STD-595.

3.16 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.17 Workmanship. The indicator shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, welding, brazing, plating, painting, riveting, machine screw assemblies, and freedom of parts from burrs and sharp edges. Mating gears shall provide a minimum of backlash necessary for smooth operation. Where necessary, antibacklash techniques shall be used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1. Certification. The contractor shall obtain FAA certification that the indicator conforms to TSO-C43 for temperature indicators.

4.2 Classification of tests. The inspection and testing of indicators shall be classified as follows:

a. Qualification tests

b. Acceptance tests.

4.3 Test conditions

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that

TABLE II

Dial and Pointer Markings

	Height or Length Inch ±0.005	Width of Line or Graduation Inch ±0.005	Color No.
Main dial numerals	0.156	0.031	37875
Main dial major graduations	0.156	0.031	37875
Subdial numerals	0.094	0.020	37875
Subdial graduations	0.062	0.020	37875
Lettering "°C x 100"	0.094	0.020	37875
"OFF"	0.094	0.020	17038
Pointers	-----	-----	37875
Background of dial	-----	-----	37038
Background of "OFF" warning	-----	-----	See 3.9

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g. High temperature and compensator lag

h. Clutch and stop effect

i. Voltage surge

4.5.2.2 Sampling plan B. Unless otherwise specified, 3 indicators shall be selected at random from the first 15 on the contract or order and subjected to the following tests as described under 4.6:

a. Sampling plan A tests

b. Sampling plan B tests of MIL-L-27160

c. Vibration

d. Low temperature exposure

e. High temperature exposure

f. Thermocouple lead resistance effect

g. Grounding of thermocouple extension wires

h. Stray signal and beat frequency toleration

i. High altitude-low temperature

j. Radio noise interference

k. Magnetic effect

l. Voltage and frequency variation

m. Thermal shock cycling

n. Reliability

o. Cycling of power OFF warning

p. Salt spray

q. Maintainability verification.

4.5.2.3 Rejection and retest. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected.

4.5.2.3.1 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 TEST METHODS

4.6.1 Examination of product. The indicator shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and markings.

4.6.2 Scale error and friction at room temperature. The millivoltage values specified in NBS Circular 561 for chromel-alumel thermocouples 0° C reference for the test points specified in table I shall be applied to the indicator. The indicator shall be lightly tapped or vibrated before each reading to remove normal instrument friction. The scale errors shall not exceed the tolerances specified in table I. Upon completion of or concurrent with the scale error test, the friction error shall be determined by slowly changing the millivolt input between test values until the next test value of millivolts is reached. When

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a test value is reached, the pointer indication shall be recorded before and after the instrument is lightly tapped or vibrated. The difference between the first and second recordings at each test point shall be recorded as the scale error when these tests are conducted concurrently. On this and subsequent tests, the power OFF warning shall be observed for proper operation. The friction error test shall be conducted while going from 0° to 1,200° C and return to 0° C. The friction error shall not exceed 2° C at any test point as viewed on the subdial.

4.6.3 Position error. Millivolts shall be applied to the indicator to produce indications of 800°, 900°, and 1,000° C. With each indication, the indicator shall be rotated clockwise and counterclockwise through an angle of 180° in each direction. The plane of the dial shall then be turned to a horizontal plane face up and face down. The millivolts shall be held constant within ±0.02 mv at each test point. The indication shall not change by more than 1° C, as viewed on the subdial, in any of the above positions at the test points specified.

4.6.4 Sealing. The indicator case shall be tested for leaks by means of a mass-spectrometer-type helium leak detector. Where a nitrogen-helium mixture is utilized as the filling medium, the detected leak rate shall not exceed 0.1 micron cubic foot per hour at a pressure differential of 1 atmosphere, dependent upon the case filling pressure. Where 100 percent helium is utilized as the filling medium, the leak rate shall not exceed 1.0 micron cubic foot per hour.

4.6.5 Warmup period. Millivoltage and 115V 400-cycle power shall be applied to the indicator to provide an indication of 0° C. The power supply shall then be turned off and the millivoltage adjusted to provide an indication of 1,000°C. The power shall then be reapplied and the time required for the pointer to reach a stable indication, within 5° C of the present 1,000° indication, shall be noted and shall not exceed 6.0 seconds. One hour shall elapse between the initial power application and the final power application to insure that the indicator is essentially at room temperature.

4.6.6 Response time. Millivoltage and power shall be applied to provide a stable indication of 700° C. The millivoltage shall then be increased to produce an instantaneous change equivalent to 350° C. The time required for the indicator to fully respond to the 350° C change shall be noted and shall not exceed 2 seconds. This procedure shall be repeated beginning at 1,050° C and instantaneously reducing the millivoltage equivalent to a change of 350° C. The same tolerance shall apply.

4.6.7 Sensitivity. Millivoltage and power shall be supplied to provide test-point indications of 800°, 900°, and 1,000° C. A change in the millivoltage equivalent to 2° C shall be made at each test point and a change of 2° ±1° C in indication shall result. The same tolerance shall apply when the input signal is either increased or decreased by 0.08 mv.

4.6.8 Low tempeature and compensator lag. The indicator shall be placed in a chamber at room temperature. Millivoltage and power shall be applied to produce a steady indication of 900° C. While maintaining the millivoltage constant within 0.02 mv and the power input within 115 ±2V 400 ±5 cps, the chamber temperature shall be reduced to -55 ±2° C at a rate of approximately 5° per minute and held at this temperature for 4 hours. While the temperature is being reduced, the indication shall not change by more than 5° C. When the

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-55° C temperature is reached, the indication shall be reduced to 0° C and the power removed until the end of the 4-hour period. The millivolt input shall then be adjusted to a value corresponding to 1,000° C, power applied, and the warmup, response time, sensitivity, and scale error tests conducted. The warmup time shall not exceed 10 seconds; the response time shall not exceed 3 seconds; the sensitivity shall be as specified in 4.6.7; and scale error tolerances shall not exceed those specified in table I for low temperature.

4.6.9 High temperature and compensator lag. The indicator shall be tested as specified in 4.6.8, except that the temperature of the chamber shall be raised to 100° C. The warmup time shall not exceed 10 seconds and the response time shall not exceed 2 seconds. The sensitivity shall be as specified in 4.6.7. The scale error shall not exceed that specified in table I for high temperature.

4.6.10 Vibration. The indicator shall be subjected to a vibration test in accordance with MIL-E-5272, procedure XII, curve B. Pointer oscillation shall not exceed 5° C double amplitude. The test point shall be 900° C with power and signal applied. Partial deflection of the POWER-OFF warning shall be cause for rejection.

4.6.10.1 Following the vibration test, the indicator shall be checked for scale error and friction as specified in 4.6.2. The scale error and friction tolerances shall not exceed those specified in table I for room temperature. No screws or other parts shall become loosened or damaged as a result of vibration.

4.6.11 Clutch and stop effect. The indicator shall be operated to indicate a temperature of 200° C. The input millivolts shall be instantaneously increased to 52 mv, maintained for 30 minutes, and reduced to provide an indication of 1,000° C. The millivoltage shall then be instantaneously reduced to -3 mv, held for 30 minutes, and returned to the original value used to produce the 200° C indication. The above procedure shall constitute 1 cycle and shall be repeated until 8 cycles have been completed. The scale error test specified in 4.6.2 shall be repeated and the scale error tolerances shall not exceed those specified in table I for scale error at room temperature. A 0° C reference junction temperature shall be maintained throughout the test. This test shall be conducted with the indicator in an ambient temperature of 50° C.

4.6.12 Voltage surge. The indicator shall be connected as specified for the voltage and frequency test and the indication set at 900° C. The voltage and frequency shall be adjusted to 115 ±1V 400 ±2 cps. A voltage surge shall be applied to the power input to provide a momentary rms voltage of 160 ±5V and return to 115 ±1V. The duration of the surge, including rise and decay time, shall be 0.09 ±0.005 second. The frequency shall then be increased to 440 ±2 cps and a voltage surge applied to provide a momentary rms voltage of 195 ±5V and return to 115 ±1V. The duration of the surge, including rise and decay time, shall be 0.02 ±0.005 second. The voltage and frequency shall then be adjusted to 115 ±1V 400 ±2 cps and accuracy tests conducted at the 800°, 900°, and 1,000° C test points. The scale error tolerances shall not exceed those specified in table I for scale error at room temperature.

4.6.13 Low temperature exposure. The indicator shall be placed in a chamber and subjected to a temperature of -65° ±2° C for 48 hours. While still at this temperature, the indicator shall be operated throughout the scale range and shall exhibit no sticking or erratic characteristics. The temperature shall then be raised to -54° ±2° C and maintained for

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24 hours. At the end of this time and while at this temperature, power shall be applied to the indicator. The warm-up time shall not exceed 10 seconds and the response time shall not exceed 3 seconds. The indicator shall then be subjected to the scale error test specified in 4.6.8. The scale error tolerances shall not exceed those specified in table I for low temperature. The indicator shall then be returned to room temperature for a minimum period of 2 hours and then subjected to a scale error test as specified in 4.6.2. The scale error tolerances shall not exceed those specified in table I for room temperature.

4.6.14 High temperature exposure. The indicator shall be placed in a chamber and subjected to a temperature of 100° ±2° C for a period of 24 hours. Power shall be applied continuously through the last 2 hours of this period. While still at this tempeature, the indicator shall be operated throughout its range and shall exhibit no sticking or erratic characteristics. The indicator shall then be subjected to and meet the requirements of the high temperature scale error test. The indicator shall then be returned to room temperature for a minimum period of 2 hours and subjected to a scale error at room temperature test as specified herein. The scale errors shall not exceed the tolerances specified in table I for scale error at room temperature. The indicator shall then be subjected to the response time test specified herein. The response time shall not exceed 2 seconds.

4.6.15 Thermocouple lead resistance effect. Power shall be applied to the indicator and the indicator subjected to a lead resistance change from 5 to 200 ohms at the 500°, 700°, 900°, and 1,000° C test points. The change in subdial pointer indication shall not exceed 2° C.

4.6.16 Grounding of thermocouple extension wires. The chromel or plus (+) wire and the alumel or minus (-) wire shall be alternately connected to the grounded or low potential side of the 115V 400-cycle power supply. Prior to the grounding of a wire, the scale error shall be determined at the 800° and 900° C test points and the response time shall be determined in the 700° to 1,050° C scale range on increasing temperature. The scale error and response tests shall be conducted with each lead alternately grounded. The scale error with either lead grounded shall not differ from the scale error without a lead grounded by more than 2°C. The response times with the leads alternately grounded shall not differ from the response times without a lead grounded by more than 1 second.

4.6.17 Stray signal and beat frequency toleration. The indicator shall be connected as shown on figure 2 and all switches opened. With the power supply to the lamp loads turned off, 115V 400-cycle power and a thermocouple signal producing an indication of 950° C shall be applied to the indicator. After 2 minutes at the 950° C indication, the cathode ray oscilloscope and the power supply to the lamp loads shall be turned on and allowed to warm up for 2 minutes. Beginning with a frequency of 380 cps, the frequency of the variable frequency supply shall be increased at a rate not exceeding 1 cps each 5 seconds until 420 cps is reached. The temperature indication shall be closely observed throughout this test. Any change in subdial pointer position or oscillation of the pointer shall be recorded and the frequency noted. This procedure shall be repeated with only the S-1 switch closed; S-1 and S-2 switches closed; S-1, S-2, and S-3 switches closed; S-1, S-2, S-3, and S-4 switches closed; and then with all switches closed. During this test, each of the thermocouple leads shall be alternately grounded with each

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NOTES:-
ALUMEL AND CHROMEL CONDUCTORS LAID PARALLEL AND CENTRALLY ARRANGED IN TIGHTLY BUNDLED CABLE FOR 20 FT. MINIMUM
SPEC. MIL-W-5846 TYPE II CLASS A
R = 100,000 ± 5,000 OHMS
L1, L2, L4 & L5 = 300 WATT LAMPS OR EQUIVALENT IMPEDANCES
L3 = 60 WATT LAMP OR EQUIVALENT IMPEDANCE

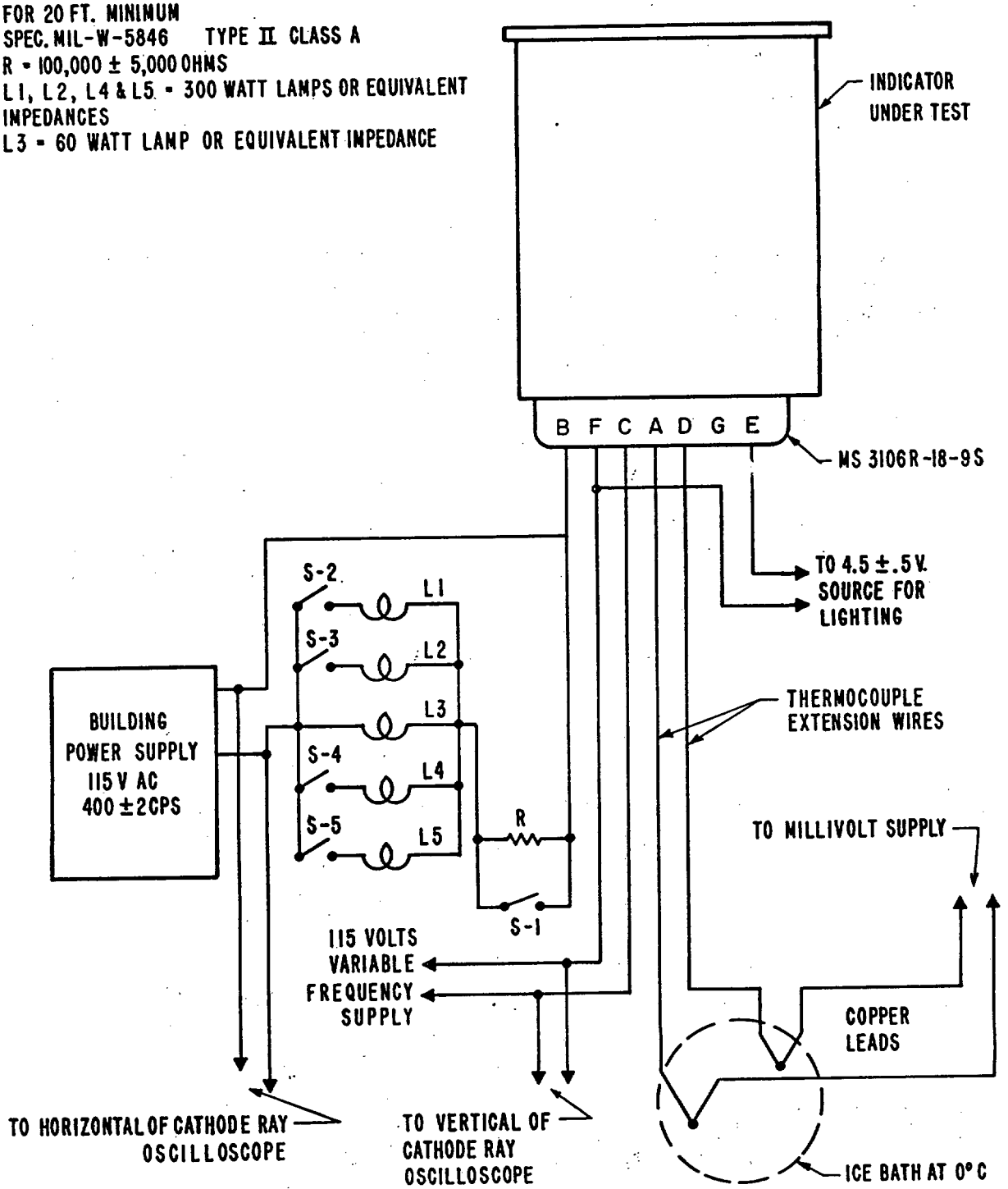


FIGURE 2. Stray Signal and Beat Frequency

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group of switches closed. Pointer oscillation shall be cause for rejection. The maximum change in pointer position throughout the test shall not exceed 4° C. If sluggishness is observed during any of the specified conditions, the response-time test shall be conducted with the worst condition maintained. The response time shall not exceed 3 seconds. With only the S-1, S-2, and S-3 switches closed, a section of the cable containing all of the conductors except the 5V lighting input conductor shall be laid against the entire length of the indicator and then across the indicator at various positions on the case length. Pointer oscillation and pointer variation shall be observed at each position of the cable as the frequency is varied from 380 to 420 cps as specified herein. Pointer oscillation shall not exceed 5° C double amplitude and pointer variation shall not exceed 4° C. The response-time test shall then be conducted with the worst condition maintained. The response time shall not exceed 3 seconds. The 115V 400 ±2 cps power supply shall be turned off and the cable shall be extended to the rear of the indicator. The integral lighting circuit shall then be supplied with 4.5 ±0.5V at the building supply frequency while the 115V frequency supply to the indicator is varied from 380 to 420 cps. Pointer oscillation shall be cause for rejection. Pointer variation shall not exceed 1° C.

4.6.18 High altitude-low temperature. The indicator shall be subjected to an absolute pressure equivalent to an altitude of approximately 100,000 feet for a period of 1 hour at a temperature of -55° ±2° C. While still at this pressure and temperature, the indicator shall be subjected to a scale error test specified in 4.6.8. The scale error tolerances shall not exceed those specified in table I for scale error at low temperature. There shall be no damage to the indicator as a result of this test.

4.6.19 Radio noise interference. Conducted susceptibility, and radiated radio noise measurements shall be made in accordance with MIL-I-6181. The conducted and radiated noises shall not exceed the values specified therein.

4.6.20 Magnetic effect. The indicator, first, not operating and second, operating on rated voltage, shall be rotated in a vertical plane about a short bar magnet compass with the nearest part of the indicator 5 ±1/2 inch from and magnetically east or west of the center of the compass. Starting directly under the compass, the indicator shall be held in position 0°, 45°, 90°, 135°, 180°, 225°, 270°, and 315° from the initial position. At each of these positions, the indicator shall be rotated about its own horizontal axis until it is in its normal viewing position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. The deflection of the compass at any of the specified positions shall not exceed 5°.

4.6.21 Voltage and frequency variation. The indicator shall be subjected to the room temperature scale error test at the 900° and 1,000° C test points only. The indicator shall then be subjected to any combination of voltages and frequencies ranging from 104 to 126V and 360 to 440 cps at the 900° and 1,000° C test points. The indication change shall not exceed 3° C.

4.6.22 Thermal shock cycling. The indicator shall be subjected to a total of 8 cycles of immersion in tap water at a temperature of 85° ±2° C and then in tap water at a temperature of 5° ±2° C. The length of time for each bath immersion shall be 30 minutes with not more than 3 seconds between immersions. No evidence of moisture penetration or damage to the inclosure

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shall result from this test. Following this test, the indicator shall be re-subjected to and shall meet the requirements of the sealing test specified in 4.6.4. Fogging of the cover glass during any part of this test shall be cause for rejection.

4.6.23 Reliability test program. Within 90 days after receipt of contract and 30 days prior to the scheduled date of initiation of testing or at some earlier date, the contractor shall submit a detailed reliability test program to the procuring activity for approval. Testing shall begin only after formal approval of the test program by the procuring activity.

4.6.23.1 Reliability test. Five sample indicators shall be tested by the contractor for reliability. Only preliminary tests at room temperature shall be conducted on these test samples prior to reliability testing. Duration of the MTBF demonstration test on each unit shall be not less than 780 hours. Acceptance, permitting production, and the total test hours shall be determined on the basis of table in MIL-R-26667 entitled "Reliability Accept-Reject Criteria (Method 2)." Recording, data handling, and reporting procedures shall also be in accordance with MIL-R-26667.

4.6.23.2 Reliability test procedure. The test equipment and fixtures shall include provisions for vibration and temperature cycling at the periodic intervals as specified herein. The vibration amplitude shall be 0.10 to 0.015 inch with a frequency of 45 to 55 cps. The amplitude and frequency chosen shall provide a vibration level of 1.5g ±10 percent. The direction of vibration is not critical. The indicators shall first be subjected to the scale-error-at-room-temperature test and then to operational cycles consisting of 4 hours of operation with power and millivoltage applied and 30 minutes with power off. The operational cycles shall be conducted at 0° C, room temperature, and 50° C in the following sequence: 4 hours at 0°, 30 minutes off; 4 hours at room temperature, 30 minutes off; 4 hours at 50° C, 30 minutes off, and 4 hours at room temperature, 30 minutes off. During the operating portion of the operational cycle, the millivoltage shall be varied to produce an indication cycle corresponding to a change ranging from 100° to 1,000° C and return to 100° C at a rate of 2.5 ±0.5 cpm. This shall constitute 1 indication cycle. The indication and operational cycles shall be repeated continuously until the indicator has been subjected to 780 hours of indication cycles. The continuity of the test may be interrupted for weekend shutdown. During each 4-hour operational cycle, the indicators shall be vibrated for a minimum period of 10 minutes each hour. At 20-hour test intervals during a 30-minute off period, one indicator shall be subjected to the scale error at room temperature test specified herein, so that all indicators are subjected to this test once each 100 hours of testing. The scale errors recorded throughout the test shall not exceed the tolerances specified in table I for scale error during MTBF testing. Upon completion of this test, the indicators shall be subjected to and shall meet the requirements of the sensitivity test specified in 4.6.7.

4.6.24 Cycling of power-off warning. This test shall be conducted prior to installation of the warning in the indicator. The power OFF warning shall be operated a total of 20,000 cycles at the rate of 15 ±2 cpm. A cycle shall consist of operating the warning from the fully off to the fully on position and return to fully off. The distance traveled by the warning mechanism shall be noted before and after testing and shall not differ by more than 1/32 inch. No malfunction shall occur during this test.

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4.6.25 Salt spray. The indicator shall be subjected to a salt spray test in accordance with MIL-E-5272, procedure I, for a period of 50 hours. The scale error at room temperature test shall be conducted immediately before and after this test. The scale error tolerance after the salt spray test shall not exceed the tolerances obtained before the test. Subsequent indicator operation shall not be affected by this test, and exposure of the base metal as a result of corrosion, including electrical receptacles, shall be cause for rejection.

4.6.26 Maintainability verification. Maintainability verification shall be in accordance with MIL-M-26512, appendix A, paragraph entitled "Maintenance Task Test and Demonstration." Random sample techniques will not be required.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A. Each indicator shall be preserved and packaged in accordance with MIL-P-7936, method III of MIL-P-116, when electrical receptacles and connections are equipped with waterproof plugs or caps and gaskets as applicable. When receptacles and connectors are not so equipped, method IA8 of MIL-P-116 will be applied.

5.1.2 Level C. Indicators shall be packaged in accordance with MIL-P-7936, method III of MIL-P-116.

5.2 Packing

5.2.1 Export. Indicators shall be packed in accordance with level A of MIL-P-7936.

5.2.2 Domestic. Indicators shall be packed in accordance with level B of MIL-P-7936.

5.2.3 Immediate use. Indicators shall be packed in accordance with level C of MIL-P-7936.

5.3 Marking. Interior and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows, as applicable:

Indicator, Temperature, Thermocouple EHU-15/A, Potentiometer Type, White Lighted

Indicator, Temperature, Thermocouple EHI-15A/A, Potentiometer Type, White Lighted.

6. NOTES

6.1 Intended use. The indicators covered by this specification are intended for use in indicating the exhaust gas and turbine inlet gas temperature of turbojet and turboprop aircraft engines.

6.1.1 The indicators can be used on instrument panels requiring case mounting in accordance with MS33639 by the application of either an instrument mounting bezel in accordance with MIL-B-6836 or a mounting clamp in accordance with MS28042.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Type of indicator required (see 1.2)

c. When sampling plan B will not be conducted

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d. Levels of packaging and packing required.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Systems Engineering Group, Attn: SEFIP, Wright-Patterson Air Force Base, Ohio and information pertaining to qualification of products may be obtained from that activity.

6.3.1 Test report. The contractor will prepare a test report in accordance with MIL-STD-831 and submit three copies of the report to the qualifying activity.

Custodian:
Air Force - 11

Preparing activity:
Air Force - 11

Review activity:
Air Force - 11, 67

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